Started on	Tuesday, 17 January 2023, 11:07 PM
State	Finished
Completed on	Tuesday, 17 January 2023, 11:09 PM
Time taken	1 min 43 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 2.00 out of 2.00

What is the most efficient method of inter-process communication?

Select one:

- a. Message Passing.
- b. Shared Memory. ✔
- c. Reading/Writing to files.
- d. Signals.

The inter-process communication mechanisms based on shared memory are the most efficient because once we get hold of a portion of shared memory, the effective communication is just working with memory. Unlike shared memory, message passing usually involves `send()` and `recv()` system calls. Signals also involve system calls, and writing and reading from a file involves working with IO, which is slow.

The correct answer is: Shared Memory.

Question **2**

Correct

Mark 2.00 out of 2.00

A Similarity between an UNIX socket and a file is:

Select one:

- a. Both UNIX sockets and files have an entry in the file system. ✔
- b. Both UNIX sockets and files are backed-up by storage (e.g disk).
- c. Both UNIX sockets and files are for persistent storage.
- d. Both UNIX sockets and files are slow inter-process communication mechanisms.

Both UNIX sockets and files have an entry in the file system. UNIX sockets are faster inter-process communication mechanisms than files. Files are usually used for persistent storage, but UNIX sockets are just IPC mechanisms, which means they cannot hold data. The content of a file is usually stored on the disk, so it's backed up by storage, whereas the UNIX socket has a buffer in Kernel Space, so it's backed up by memory.

The correct answer is: Both UNIX sockets and files have an entry in the file system.

Question **3**

Correct

Mark 2.00 out of 2.00

UNIX sockets, pipes and FIFOs are:

Select one:

a. inter-process communication mechanisms based on shared memory in user space.

b. inter-process communication mechanisms based on message transfer that use buffers in kernel space and are accessed through file descriptors. ✔

c. inter-process communication mechanisms based on message transfer that use buffers in user space and are accessed through file descriptors.

d. inter-process communication mechanisms that do not use memory.

UNIX sockets, pipes and FIFOs are inter-process communication mechanisms based on message transfer. The underlying implementation is a buffer in kernel space. So all of them use some kind of shared memory, but not in user space. Accessing all of them is done through file descriptors.

The correct answer is: inter-process communication mechanisms based on message transfer that use buffers in kernel space and are accessed through file descriptors.

Question **4**

Correct

Mark 2.00 out of 2.00

For a local communication what is better to use between UNIX and BSD sockets?

Select one:

a. UNIX sockets because we can't use BSD sockets locally.

b. UNIX sockets because they are faster than BSD sockets. ✔

c. BSD sockets because they use the loopback interface and UNIX sockets use files.

d. BSD sockets because they are faster than UNIX sockets.

For local communication it is better to use UNIX sockets because they are faster than BSD. UNIX sockets use some shared buffers in kernel space, while BSD sockets use the loopback interface for local communication.

The correct answer is: UNIX sockets because they are faster than BSD sockets.

Question **5**

Correct

Mark 2.00 out of 2.00

Can we use BSD sockets for local communication?

Select one:

- a. No. Because BSD sockets are network sockets and can only be used for remote communication.
- b. Yes. Using BSD sockets bound to local files.
- c. Yes. Using any network interface. ✔
- d. Yes. Only on port 80.

Although BSD sockets are used for network communication, they can also be used locally for communication between processes of the same machine using any local interface (e.g. the loopback one).

The correct answer is: Yes. Using any network interface.

◄ Curs 11: App Interaction (part 1): Recap

Jump to...

Curs 12: App Interaction (part 2): Recap ►